pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Population Stability and Drift

WEEK 2 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: has the population moved?

Two populations

1.7173

10 Yr level PSI

996

new firms, 11 to 14

The curve's 2019-2022 against 2015-2018 (real); the cohort's entrants against years 1 to 10 (synthetic).

Today's question

Has the population a model sees moved from the one it was built on?

PSI answers with one number per variable.

The held-out years stay closed. The reference and its bins are fixed before the compared sample is read.

Drift is described

A PSI says how far a sample sits from its reference, measured one way. It never says where a rate or a population goes next.

Allowed names

PSI: how far the mix moved

a shift in the inputs

a description of two samples

Never called

a forecast that levels return

a view on rates

a signal to trade

What to do about a drifted input is a question for whoever owns the model.

How PSI works

The reference	Fixed first: the population a model was built on
The bins	Cut at the reference's own quantiles: ten bins, a tenth each (psi_edges)
The shares	The share of each sample in each of the same bins
The terms	(compared - reference) x ln(compared / reference): never negative
The PSI	The sum: 0 when the shares agree, growing as they move apart (psi)

PSI by hand

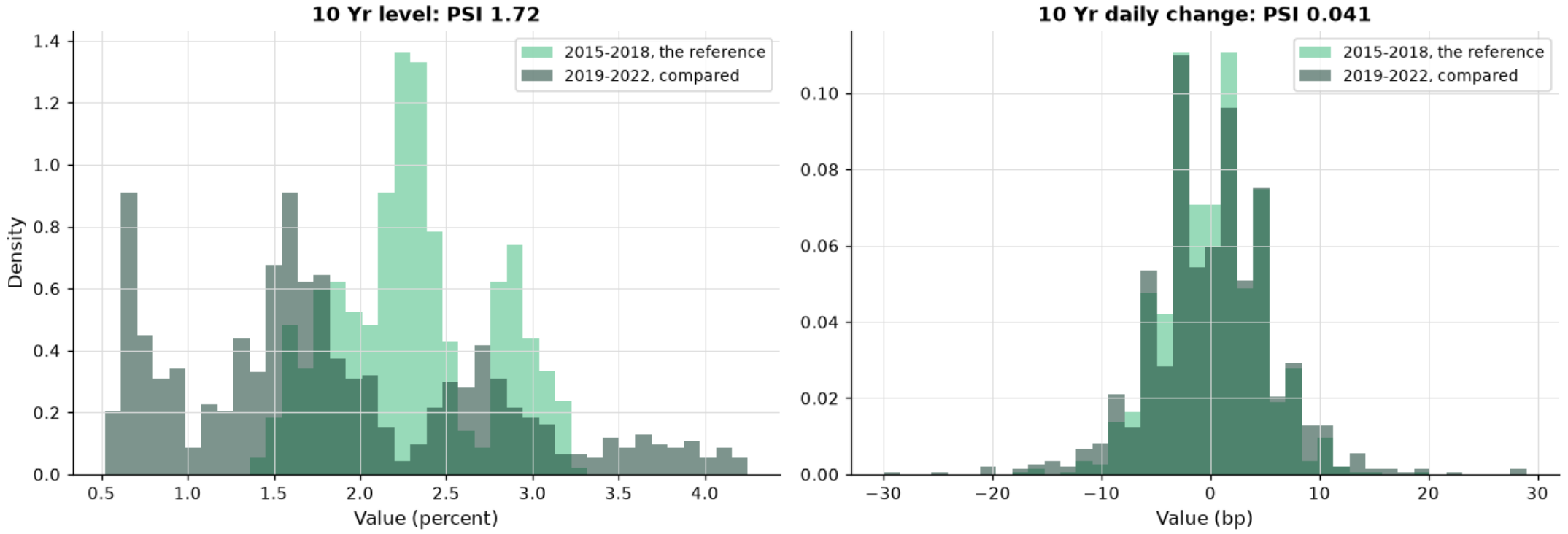
Bin	From	To	Reference	Compared	Term
1	-inf	1.740	0.105	0.540	0.713
2	1.740	1.890	0.097	0.081	0.003
3	1.890	2.090	0.099	0.064	0.015
4	2.090	2.200	0.105	0.016	0.168
5	2.200	2.280	0.106	0.003	0.367
6	2.280	2.360	0.102	0.007	0.255
7	2.360	2.450	0.089	0.020	0.103
8	2.450	2.820	0.103	0.115	0.001
9	2.820	2.940	0.098	0.029	0.084
10	2.940	inf	0.096	0.125	0.008

1.7173

PSI, 10 Yr level

2019-2022 against the deciles of 2015-2018, in percent. More than half the compared days sit in the lowest bin.

The level against its change



A level wanders (PSI 1.7173); its daily change barely does (PSI 0.0408).

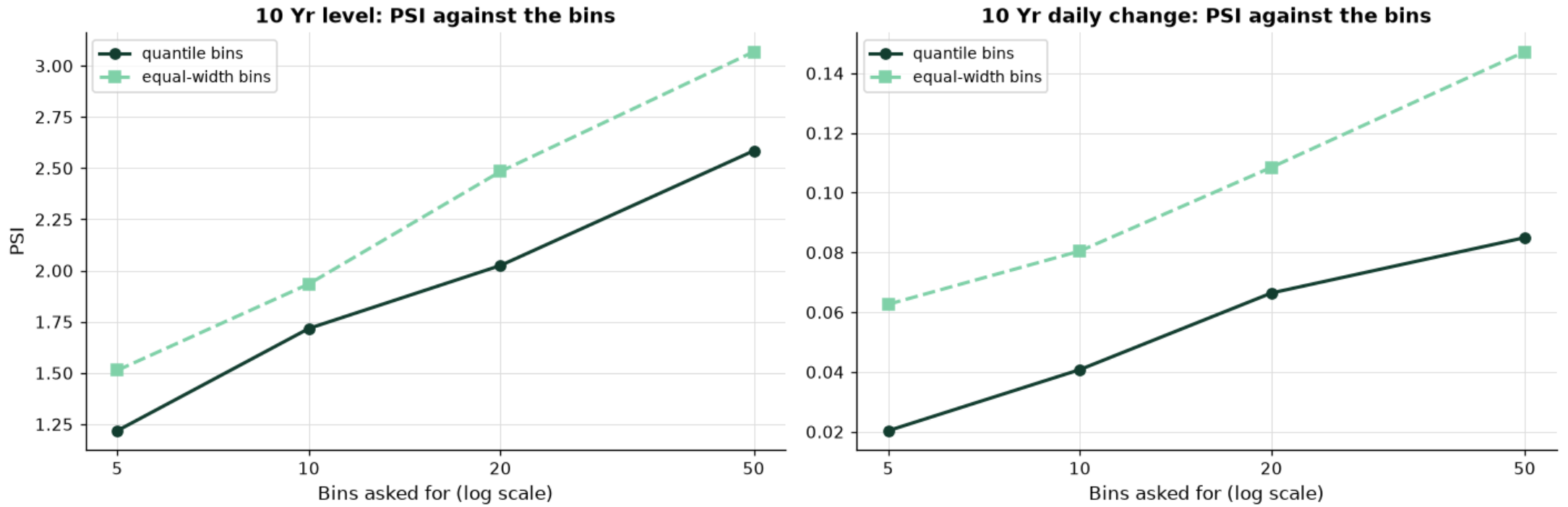
The empty bin and its floor

Floor	PSI, one quarter	PSI, 2019-2022
1e-06	12.3173	1.7173
0.0001	8.1895	1.7173
0.001	6.0935	1.7173
0.01	3.8675	1.5357

All 61 days of the first quarter of 2021 sit in one reference bin; nine bins are empty. ln(0) has no value, so a floor raises an empty share before the log.

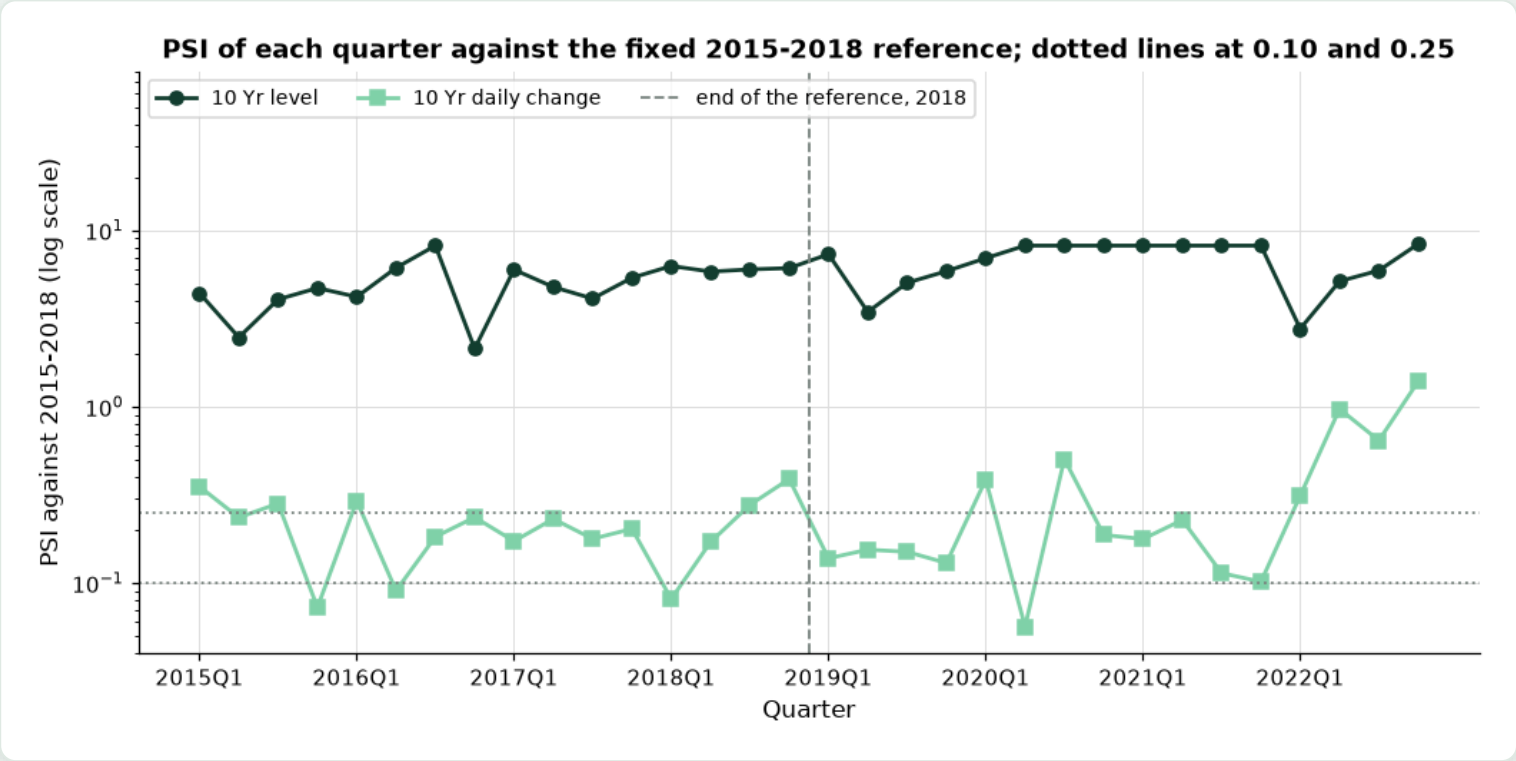
The quarter's PSI is the floor's choice. Read it as "outside the reference", not as a size.

How many bins, and how they are cut



Every line climbs. Fix the bins before any comparison; ten quantile bins is the common choice and today's.

PSI quarter by quarter



0.217

inside, median

0.183

after, median

Daily change. About 60 days in ten bins: noise is large even inside the reference. The level's median inside it is 5.08.

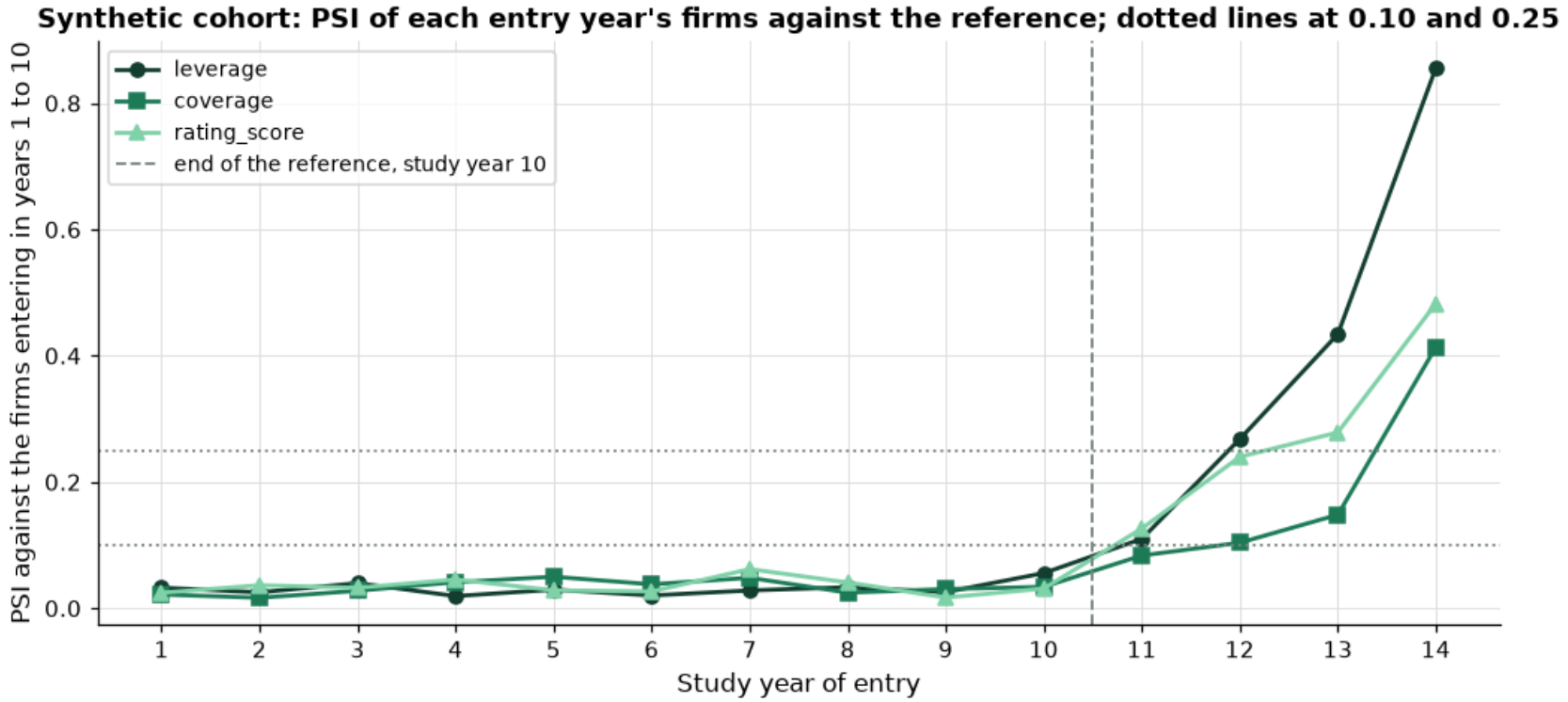
The cohort's inputs

Input	PSI	Bins	Median, years 1-10	Median, years 11-14
leverage	0.333	10	4.125	4.945
rating_score	0.232	10	10.000	12.000
coverage	0.122	10	5.855	4.720

2,508 synthetic firms entering in study years 1 to 10 against 996 entering in 11 to 14 (psi_table). The generator's shift toward more leverage and a lower rating is written in DATA.md.

A PSI has no direction; the medians say which way.

Entry year by entry year



Years 1 to 10 show the noise of about 250 firms a year. From year 11 the planted shift appears, gradually.

PSI, KS, and Jensen-Shannon

Comparison	PSI	KS statistic	KS p-value	Jensen-Shannon
10 Yr level, 2019-2022	1.72	0.437	9.56e-87	0.509
10 Yr daily change, 2019-2022	0.0408	0.0419	0.33	0.0856
10 Yr level, 2021 Q1	8.19	0.895	2.85e-54	0.866
leverage, entering 11-14	0.333	0.235	6.69e-35	0.241
coverage, entering 11-14	0.122	0.142	5.5e-13	0.147
rating score, entering 11-14	0.232	0.181	5.57e-21	0.202

The KS p-value measures evidence, not size. KS and Jensen-Shannon stop at 1. PSI sees nothing inside a bin.

Rules of thumb without error rates

Bilal Yurdakul, PhD dissertation, Western Michigan University, 2018

"Statistical Properties of Population Stability Index"

Rules of thumb in use: below 0.10 "little shift"; 0.25 marks "significant shift, action required"

"these benchmarks are being used without reference to Type I or Type II error rates." The notebook draws the two lines and decides nothing with them.

Source: scholarworks.wmich.edu/dissertations/3208, the abstract. Its middle band does not display legibly there, so it is not quoted.

Excel side by side

Excel side by side (tested in Excel)

`=PERCENTILE.INC(A2:A1001,{0.1;0.2;0.3;0.4;0.5;0.6;0.7;0.8;0.9})`	edges 1.74 to 2.94
`=FREQUENCY(B2:B1002,E2:E10)/COUNT(B2:B1002)`	lowest bin 0.5405
`=SUMPRODUCT((G2:G11-F2:F11)*LN(G2:G11/F2:F11))`	PSI 1.7173
`=IF(H2#<0.0001,0.0001,H2#)`	the floor
`=SUMPRODUCT((I2:I11-F2:F11)*LN(I2:I11/F2:F11))`	quarter 8.1895

FREQUENCY counts up to each edge, as psi does. Without the floor, LN(0) returns #NUM!. Every value matched Python.

Save the day: inside one commit

```
git show --stat HEAD
```

The last commit's message and the files it changed, with the lines added in each.

A quick check that a commit holds what its message says, and nothing else.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: edges cut from the compared sample: 1.0791 on a hand case worth 3.7970.

Code that runs is not the same as code that is right.

What you can do now

- PSI by hand and with psi_edges and psi
- A level drifts; its daily change barely does
- An empty bin, its floor, and the bins fixed first
- PSI window by window, and per input with psi_table
- PSI beside KS and Jensen-Shannon; 33 tests pass

Week 2, Day 4 Case study: monitoring a credit model